

BANATSKI KARLOVAC, Serbia

WMO: 131800

Lat: 45.050N

Lon: 21.033E

Elev: 299

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	17.4	7.7	8.3	15.0	13.3	11.0	19.6	36.5	33.6	32.6	36.7	5.7	340	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	92.7	70.4	89.9	69.9	87.0	68.7	74.0	85.3	72.2	84.2	70.8	82.6	6.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.8	114.6	79.0	68.8	107.1	76.4	67.3	101.3	74.4	37.7	84.9	36.1	84.3	34.8	82.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.1	23.4	19.7	DB	5.0	96.4	8.0	3.3	-0.8	98.8	-5.5	100.7	-10.0	102.5	-15.8	104.9	(4)
(5)			WB	4.6	76.9	7.8	2.6	-1.0	78.8	-5.6	80.3	-9.9	81.8	-15.6	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	33.6	37.0	45.1	55.2	63.6	69.3	72.4	72.6	64.1	54.4	45.6	35.6	(6)	
	DBStd	15.85	9.10	9.73	8.59	7.31	6.74	6.25	5.35	5.62	6.82	8.20	8.56	8.63	(7)	
	HDD50	1798	510	372	197	34	1	0	0	0	0	52	184	448	(8)	
	HDD65	4756	974	785	616	302	109	30	7	7	97	337	582	910	(9)	
	CDD50	3307	1	7	46	190	423	579	693	701	424	188	52	3	(10)	
	CDD65	789	0	0	0	8	65	158	235	244	71	8	0	0	(11)	
	CDH74	8089	0	0	2	100	617	1620	2480	2514	677	79	0	0	(12)	
	CDH80	3042	0	0	0	18	132	575	1023	1085	203	6	0	0	(13)	
(14) Wind	WSAvg	8.0	9.0	9.3	9.7	9.7	8.0	6.5	5.8	5.7	6.7	7.8	9.5	8.6	(14)	
(15) Precipitation	PrecAvg	25.2	1.5	1.5	1.5	2.0	2.5	3.0	3.1	2.2	2.4	2.0	1.8	1.9	(15)	
	PrecMax	36.9	3.0	3.0	3.9	4.3	5.0	6.8	10.5	7.8	6.9	5.0	5.0	5.0	(16)	
	PrecMin	12.2	0.3	0.1	0.3	0.1	0.8	0.7	0.1	0.1	0.2	0.0	0.0	0.2	(17)	
	PrecStd	5.7	0.8	0.8	0.9	1.1	1.1	1.6	2.4	1.7	1.5	1.3	1.1	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	64.0	71.9	82.6	87.2	93.1	96.6	97.7	91.3	80.3	70.2	57.4	(19)	
		MCWB	49.2	50.7	54.0	61.3	65.4	70.8	71.2	70.8	67.5	63.0	57.0	51.1	(20)	
	2%	DB	51.6	58.2	67.3	76.5	83.6	89.9	92.2	93.4	85.4	76.2	65.4	53.5	(21)	
		MCWB	46.6	49.1	52.2	58.2	64.3	70.3	71.0	70.0	65.5	61.1	55.5	48.8	(22)	
	5%	DB	48.4	53.7	62.8	72.3	80.7	86.6	89.3	89.9	81.0	72.1	61.3	50.0	(23)	
		MCWB	45.0	47.3	50.5	56.3	63.3	69.1	70.4	69.7	64.2	59.1	53.5	46.3	(24)	
	10%	DB	45.3	49.8	58.5	68.5	77.4	83.3	86.1	86.4	77.2	67.8	57.7	46.7	(25)	
		MCWB	42.8	45.0	48.7	54.7	62.5	67.7	69.2	68.3	62.5	57.9	51.7	44.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	53.0	56.2	62.7	69.1	76.1	76.4	76.4	70.0	64.7	58.8	52.8	(27)	
		MCDB	54.3	60.4	67.4	75.5	79.3	85.7	86.8	86.6	83.8	74.2	67.1	55.8	(28)	
	2%	WB	47.5	50.1	53.6	60.0	67.1	73.1	73.9	73.6	67.7	62.6	56.6	48.8	(29)	
		MCDB	51.0	56.8	63.5	72.1	77.6	83.5	85.7	85.2	80.1	73.2	63.4	52.6	(30)	
	5%	WB	45.1	47.7	51.7	58.2	65.6	71.1	72.2	71.6	65.8	60.7	54.4	46.6	(31)	
		MCDB	48.1	53.0	60.9	68.7	75.7	82.3	84.4	84.2	76.6	69.1	59.9	49.5	(32)	
	10%	WB	43.0	45.4	49.8	56.4	64.0	69.3	70.9	70.0	64.3	59.0	52.3	44.2	(33)	
		MCDB	45.2	49.6	57.4	65.6	73.6	79.2	82.6	82.3	73.8	66.8	57.3	46.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	12.6	16.8	18.5	19.3	20.2	21.8	22.4	19.6	18.2	13.4	9.7	(35)	
		MCDBR	14.4	19.3	23.4	25.1	24.6	26.0	27.5	28.4	26.1	24.2	18.5	14.2	(36)	
	5% WB	MCWBR	10.0	11.3	11.9	10.8	8.9	9.6	10.1	9.5	9.7	10.9	9.4	9.7	(37)	
		MCWBR	13.4	17.0	20.0	21.1	21.3	23.0	24.0	23.8	21.9	21.4	17.1	13.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.371	0.404	0.460	0.447	0.441	0.447	0.459	0.434	0.396	0.361	0.343	(40)	
		taud	2.328	2.245	2.208	2.065	2.166	2.219	2.198	2.165	2.202	2.301	2.352	2.352	(41)	
	Ebn at Noon		237	252	260	254	262	264	260	252	247	240	229	225	(42)	
		Edh at Noon	26	33	39	49	46	44	44	44	39	31	25	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		385	650	1045	1450	1759	1953	1966	1748	1221	835	504	319	(44)	
	RadStd		47	87	130	145	167	179	151	166	132	93	44	46	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.10	N/A	N/A	N/A	+1.15	+1.37	N/A	N/A	N/A	N/A	(46)
(47)	Regional (59 neighbors)	+1.21	N/A	N/A	N/A	+0.67	N/A	-200	-311	+212	+115	(47)

Nomenclature: See separate page